

Slide 1: Title Slide

Title: Spam Email Detection Using Machine Learning

Subtitle: Mini Project

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Slide 2: Introduction

- Email spam is unwanted or harmful messages sent in bulk.
 - It can contain advertisements, phishing links, or malware.
 - Detecting spam emails helps improve security and user experience.
 - Machine Learning can automatically classify emails as spam or not spam.
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Slide 3: Objective

- To build a model that classifies emails into:
 - Spam (1)
 - Not Spam / Ham (0)
 - Use Machine Learning techniques for text classification.
 - Provide real-time prediction for user input emails.
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Slide 4: Technologies Used

- Python
 - Pandas (Data handling)
 - Scikit-learn (ML model)
 - Matplotlib (Visualization)
 - CountVectorizer (Text to Numeric)
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Slide 5: Dataset

- Dataset Name: Emails_data.csv
- Contains two columns:
 - Message (Email text)
 - Label (spam / ham)

- Label conversion:
 - Spam → 1
 - Ham → 0
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Slide 6: Data Preprocessing

- Convert labels into numeric values.
 - Use CountVectorizer to convert text into numerical format.
 - Each word is converted into frequency values.
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Slide 7: Feature Extraction

- CountVectorizer is used to transform text data.
 - Converts email text into a matrix of token counts.
 - Example:
 - “Free offer” → [1, 1, 0, ...]
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Slide 8: Data Visualization

- Scatter plot used for visualization.
 - X-axis: Word frequency
 - Y-axis: Label (0 = Ham, 1 = Spam)
 - Helps understand data distribution.
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Slide 9: Train-Test Split

- Dataset divided into:
 - Training Data (80%)
 - Testing Data (20%)
 - Ensures model is evaluated on unseen data.
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Slide 10: Model Used

- Algorithm: Multinomial Naive Bayes
- Suitable for text classification problems.
- Works based on probability.

Slide 11: Model Training

- Train model using training data.
- Model learns patterns of spam and non-spam emails.

Slide 12: Prediction

- Model predicts labels for test data.
- Output: Spam or Not Spam

Slide 13: Accuracy Evaluation

- Accuracy Score is used to measure performance.
- Formula:
$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Predictions}}$$

Slide 14: User Input Testing

- User can enter any email text.
- Model predicts:
 - Spam Email
 - Not Spam Email
- Loop continues until user types "exit".

Slide 15: Advantages

- Fast and automatic detection
- Reduces manual effort
- Improves email security

Slide 16: Limitations

- Accuracy depends on dataset quality
 - Cannot detect new types of spam easily
 - Needs proper training data
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Slide 17: Future Scope

- Use advanced models like Deep Learning
 - Improve accuracy with large datasets
 - Add GUI for better user experience
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Slide 18: Conclusion

- Successfully built a spam detection system
 - Used Machine Learning for classification
 - Model can predict spam emails effectively
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Slide 19: Thank You

Thank You!

Any Questions?